

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Xe-133			FORMS: Gaseous, soluble		
PHYSICAL CHARACTERISTICS HALF-LIFE: 5.243 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
81	38	101	99		
31	41	45*	55		
35	8	75*	7		
36	1	25*	4		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 200 mCi B – level (moderate hazard): > 200 mCi, ≤ 10 Ci A – level (high hazard): > 10 Ci					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 0.38 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 66 mrem/h Contamination skin dose, uniform deposit of 1 µCi per cm²: N/A; GAS			INTERNAL RADIATION HAZARDS Derived Air Concentration: 0.1 µCi/l The value above indicates the air-concentration of radioactivity that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE) after 2000 hours of exposure (one working year).		
SHIELDING Gammas/X-rays: 0.4 mm of lead will reduce the gamma dose rate by 90%. Betas/electrons: 0.8 mm of plastic will absorb all emissions. Bremsstrahlung may be created and require additional shielding.			DOSIMETRY AND BIOASSAY REQs Whole-body and finger-ring dosimeters are required for handling 5 mCi or more, or 1 mCi amounts weekly .		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: NaI, PGM 2. Xe-133 is heavier than air and difficult to be kept in solution form. One should work in well-ventilated areas. 3. Syringes containing Xe133 should be regarded as leakage and radiation sources, even after being emptied. 4. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 5. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required. 6. Segregate waste to those with half-lives of between 1 and 8 days. Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot. 7. Limit soluble waste to the sewer to less than 1000 µCi/day per lab.					

References:

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- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
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